

PROJECT DOCUMENTATION

Pet Adoption & Rescue Management System

A Database-Driven Platform Connecting Adopters and Rescue Shelters

Prepared by the Development Team - RescueOps

Team Leader: Sarthak

Team Members: Shravani • Kadambari • Avishkar • Nishtha • Shaun

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1. Project Overview

1.1 Project Title

Pet Adoption & Rescue Management System

1.2 Objectives

To develop a real-time platform that connects prospective adopters with rescue shelters in order to streamline the adoption process. The system is driven by a social welfare goal to reduce stray populations and promote responsible pet ownership. It efficiently manages pet inventory, medical histories, and secure adoption transactions, while providing administrators with actionable analytics to maximize community impact.

1.3 Scope of the System

- Dual role-based access for Adopters and Administrators, each with a dedicated dashboard.
- Centralized, real-time pet inventory covering identity, health, and shelter-assignment data.
- Searchable and filterable public catalogue of pets available for adoption.
- End-to-end vaccination and medical history tracking for every pet in the system.
- A controlled adoption transaction workflow with validation and duplication checks.
- An administrative analytics dashboard with tabular reports and Matplotlib visualizations.

2. Team & Module Responsibilities

The project is divided into six functional modules. Each module has a single owner responsible for its design, implementation, and testing. The team leader additionally oversees the security and access-control logic that governs how the other five modules are exposed to each user role.

2.1 Team Roster

Name	Role	Module Owned	Module Title
Sarthak	Team Leader	Module 1	Authentication & Role Management
Shravani	Team Member	Module 2	Pet Inventory & Listing Management
Kadambari	Team Member	Module 3	Pet Search & Discovery
Avishkar	Team Member	Module 4	Vaccination & Medical Tracking
Nishtha	Team Member	Module 5	Adoption Transaction Management
Shaun	Team Member	Module 6	Administrative Dashboard & Analytics

2.2 Detailed Responsibilities

Sarthak — Module 1: Authentication & Role Management

Role: Team Leader

Split-login routing, adopter/administrator registration, security implementation (password handling, recovery), data-migration logic for elevating normal users to Administrators, direct admin creation, and configuring module-level access-control rules to keep Adopter and Admin capabilities properly separated.

Shravani — Module 2: Pet Inventory & Listing Management

Role: Team Member

CRUD operations on pet records, shelter assignment logic, health-status and media updates, and real-time management of pet availability status (Available / Pending / Adopted).

Kadambari — Module 3: Pet Search & Discovery

Role: Team Member

SQL filtering logic for keyword and category search, species/age/location category browsing, and formatting of user-facing search results and detailed pet previews.

Avishkar — Module 4: Vaccination & Medical Tracking

Role: Team Member

One-to-many relationship logic linking pets to their medical histories, vaccination record entry and edits, veterinarian logging, and tracking of upcoming/overdue vaccination due dates.

Nishtha — Module 5: Adoption Transaction Management

Role: Team Member

SQL transaction logic to link adopters and pets, availability and duplicate-adoption checks, status updates on successful adoption, and handling of pet returns.

Shaun — Module 6: Administrative Dashboard & Analytics

Role: Team Member

System monitoring views, generation of operational and medical reports, and creation of Matplotlib visualizations for adoption trends, species distribution, and shelter occupancy.

3. Functional Modules

The following sections describe each module's functional requirements in full detail, including the specific access level granted to each user role.

Module 1: Authentication & Role Management

Access Level: Admin (Full Access) | Adopter (Self-Service Access) — Owner: Sarthak (Team Leader)

1.1 System Entry & Role Selection

- Initial Prompt: Present the user with a choice at startup to either "Log in as Adopter" or "Log in as Administrator".
- Routing: Direct the user to the corresponding authentication flow based on their selection.

1.2 Adopter Registration & Account Dashboard

- Account Creation: Capture Full Name, Email Address, Phone Number, and Password to insert a new record into the Adopters table.
- Account Dashboard: Provide a centralized dashboard for normal users to manage their personal profile, update contact info, and track their active adoption requests.
- Account Security: Provide functions to change passwords or recover forgotten passwords.

1.3 Credential Verification & Session Management

- Role-Specific Login: Verify login credentials against the specific database table chosen at startup (Adopters table or Administrators table).
- Session Access: Grant access to the Admin Dashboard or Adopter Dashboard upon successful validation, and handle logout functionality.

1.4 Administrator Privilege Management

- Account Elevation (Admin Only): Allow an existing administrator to search for a normal user and grant them Admin privileges.
- Data Migration: Program the logic to migrate the user's data from the Adopters table into the Administrators table when elevated.
- Direct Admin Creation: Allow existing admins to directly register a new administrative colleague without them needing a normal user account first.

1.5 Module Access & Authorization Configuration

- Access Control Logic: Define and implement the rules dictating which modules (1-6) are accessible based on the authenticated role.
- Dynamic Menu Rendering: Ensure the system UI only displays options and modules authorized for the active user's role.
- Security Validation: Prevent forced or unauthorized access to restricted administrative functions (Modules 2, 4, 6) by normal users.

Module 2: Pet Inventory & Listing Management

Access Level: Admin (Full Access) | Adopter (Read-Only Access) — Owner: Shravani

2.1 Add New Pet

- Basic Information: Enter Pet Name, Species, Breed, Age, Gender.
- Shelter Assignment: Assign to specific Shelter ID.

2.2 Pet Details Management

- Health & Condition: Update Health Status (Healthy, Needs Care, Special Needs).
- Media: Upload Pet Photo (Optional).

2.3 Status & Availability Updates

- Status Categories: Available, Pending, Adopted.
- Real-Time Updates: Change status instantly when a transaction occurs.

2.4 Edit & Delete Listings

- Modify Records: Update Age, Update Health Status.
- Remove Records: Delete Pet Listing, Confirm Deletion.

Module 3: Pet Search & Discovery

Access Level: Admin (Full Access) | Adopter (Available Pets Only) — Owner: Kadambari

3.1 Keyword Search

- Search by Name: Enter pet name to display matching results.
- Search by Breed: Enter breed to display specific dogs/cats.

3.2 Category & Filter Search

- Species Filter: Browse by Dog, Cat, Bird, etc.
- Age Filter: Filter by Puppy/Kitten, Adult, Senior.
- Location Filter: Search by Shelter Location/City.

3.3 Search Results & Preview

- Results Display: Display matching pets with basic info.
- Detailed Preview: View complete pet medical and shelter history.

Module 4: Vaccination & Medical Tracking

Access Level: Admin (Full Access) | Adopter (Read-Only Access) — Owner: Avishkar

4.1 Medical Record Entry

- Vaccination Details: Select Pet ID, Enter Vaccine Name.
- Date Management: Record Administration Date, Set Next Due Date.

4.2 Medical History Management

- Record Updates: Edit existing vaccination dates or details.

- Vet Details: Log the Veterinarian who administered the care.

4.3 Alerts & Tracking

- Due Date Tracking: Identify pets with upcoming vaccination requirements.
- Status Flags: Flag pets that are medically ineligible for adoption.

Module 5: Adoption Transaction Management

Access Level: Admin (Process & Approve) | Adopter (Request & View History) — Owner: Nishtha

5.1 Transaction Processing

- Link Entities: Link Adopter ID to Pet ID.
- Log Transaction: Record Adoption Date, Update Pet Status to "Adopted".

5.2 Validation & Duplication Prevention

- Availability Check: Ensure pet status is "Available" before processing.
- Duplicate Check: Prevent multiple adoptions for the same pet.

5.3 Return Management

- Process Returns: Log Return Date, Revert Pet Status to "Available".
- History Tracking: Maintain record of the failed adoption without deleting the history.

Module 6: Administrative Dashboard & Analytics

Access Level: Admin (Full Access) | Adopter (No Access) — Owner: Shaun

6.1 System Monitoring

- User Records: View all registered adopters and admins.
- Transaction Records: View completed adoptions and pending requests.

6.2 Report Generation

- Operational Reports: Generate active available pets report.
- Medical Reports: Generate list of overdue vaccinations.

6.3 Graphical Analytics (Matplotlib)

- Adoption Trends: Plot line graphs showing monthly adoption trends.
- Distribution: Display pie charts for pet species distribution (e.g., % Dogs vs % Cats).
- Occupancy: Bar charts showing shelter capacity vs. current occupancy.

4. System Workflow

Phase 1: Entry & Authentication (Module 1 — Sarthak)

- System Launch: The user launches the application and is asked to select their account type (Adopter or Administrator).
- Registration/Login: New users register and their data is saved to the Adopters table; existing users log in with their email and password, verified against their selected role table.
- Role Routing: Admin accounts are routed to the Admin Dashboard, granting access to Modules 1-6. Adopter accounts are routed to the Adopter Dashboard, with dynamically restricted access limited to view capabilities in Modules 2, 3, and 4, and limited request capabilities in Module 5.

Phase 2: Administrative Setup & Inventory (Modules 2 & 4 — Shravani & Avishkar)

- Adding Inventory: The Admin adds a new rescued pet into the system (Name, Breed, Age) and assigns it to a specific Shelter_ID. The pet's initial status is set to "Available".
- Medical Logging: The Admin immediately navigates to the Medical module to log the pet's initial vaccinations and sets the "Next Due Date".
- Real-Time Availability: The pet is now live in the database and visible to all users.

Phase 3: The Adopter Journey (Module 3 — Kadambari)

- Discovery: The Adopter logs in and opens the Search & Filter engine.
- Filtering: They filter the database to find "Available" pets (e.g., searching for "Cats" that are "Kittens" in "City X").
- Previewing: The Adopter selects a specific pet to view its complete profile, including its medical history and current shelter location (Read-Only access).
- Initiating Adoption: The Adopter clicks "Request Adoption" for that specific pet.

Phase 4: Transaction & Status Update (Module 5 — Nishtha)

- Admin Review: The Admin sees the pending adoption request on their dashboard.
- Processing the Transaction: The Admin approves the adoption.
- Database Execution (Real-Time): The system logs the Adopter_ID, Pet_ID, and the current date into the Adoption_Transactions table. Simultaneously, the system updates the Pets table, automatically changing the pet's status from "Available" to "Adopted".
- Visibility Change: Because the status is now "Adopted", the pet instantly disappears from the public search results for other adopters.

Phase 5: Analytics & System Monitoring (Module 6 — Shaun)

- Data Aggregation: The Admin opens the Analytics module.
- Report Generation: The system runs complex SQL JOIN queries across the tables.
- Visualization: The system uses Python (Matplotlib) to generate real-time visual charts based on the database, such as monthly adoption trend lines or shelter capacity bar charts.

5. Database Design

The system is backed by six relational tables. Primary and foreign keys enforce referential integrity between adopters, administrators, pets, shelters, vaccinations, and adoption transactions.

5.1 Adopters (Normal Users)

Field	Description
Adopter_ID	Primary Key
Full_Name	Adopter's full name
Email	Unique login identifier
Password_Hash	Securely hashed password
Phone_Number	Contact number
Account_Created_Date	Date the account was created

5.2 Administrators

Field	Description
Admin_ID	Primary Key
Full_Name	Administrator's full name
Email	Unique login identifier
Password_Hash	Securely hashed password
Phone_Number	Contact number
Granted_By_Admin_ID	Tracks which Admin granted the privilege

5.3 Pets

Field	Description
Pet_ID	Primary Key
Name	Pet's name
Species	Dog, Cat, Bird, etc.
Breed	Pet's breed
Age	Pet's age

Field	Description
Health_Status	Healthy / Needs Care / Special Needs
Adoption_Status	Available / Pending / Adopted
Shelter_ID	Foreign Key → Shelters

5.4 Shelters

Field	Description
Shelter_ID	Primary Key
Shelter_Name	Name of the shelter
Location	City / area of the shelter
Capacity	Maximum number of pets the shelter can house

5.5 Vaccinations

Field	Description
Vaccination_ID	Primary Key
Pet_ID	Foreign Key → Pets
Vaccine_Name	Name of the vaccine administered
Date_Administered	Date the vaccine was given
Next_Due_Date	Date the next dose is due
Veterinarian	Vet who administered the care

5.6 Adoption_Transactions

Field	Description
Transaction_ID	Primary Key
Pet_ID	Foreign Key → Pets
Adopter_ID	Foreign Key → Adopters
Adoption_Date	Date the adoption was finalized
Return_Date	Date of return, if applicable

6. Summary

This document consolidates the full functional specification of the Real-Time Pet Adoption & Rescue Management System: its objectives, six core modules, end-to-end system workflow, database schema, and the division of responsibilities across the team. Each module owner is accountable for the design, build, and testing of their respective area, while the Team Leader additionally coordinates the authentication and access-control logic that ties every module together into a single, role-aware system.

Module	Title	Owner
1	Authentication & Role Management	Sarthak (Team Leader)
2	Pet Inventory & Listing Management	Shravani
3	Pet Search & Discovery	Kadambari
4	Vaccination & Medical Tracking	Avishkar
5	Adoption Transaction Management	Nishtha
6	Administrative Dashboard & Analytics	Shaun